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SECOND REVIEW OF THE PROGRAM UNDER THE EXTENDED CREDIT FACILITY, REQUEST FOR WAIVERS OF NONOBSERVANCE OF PERFORMANCE CRITERIA, AND FINANCING ASSURANCES REVIEW—DEBT SUSTAINABILITY ANALYSIS

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Following the restructuring of the debt to Glencore and the progress made in clearing external arrears, debt vulnerabilities declined significantly and the external risk rating has been upgraded to high. The Debt Sustainability Analysis[1] (DSA) shows that all debt burden indicators, except the debt-service-to-revenue ratio which has minor and temporary breaches, are below their respective thresholds in the baseline from 2018 onwards. The debt-service-to-revenue ratio, falls below the threshold in 2019 and remains so throughout the projection period, except for minor breaches in 2020 and 2021. Overall, total public debt vulnerabilities are elevated although the present value (PV) of the public debt-to-GDP ratio remains on a downward trajectory. The fixed primary balance scenario, which keeps the primary deficit-to-GDP ratio unchanged from 2017, shows the debt ratio declining at a slower pace throughout the forecast period, further highlighting the need to adhere to the prudent fiscal policy framework underpinning the IMF-supported program. Adoption and implementation of an appropriate debt management strategy, while making progress in economic diversification would further reduce vulnerabilities.

[1] Chad's three-year average score of the Country Policy and Institutional Assessment (CPIA) for 2015–17 is estimated at 2.60. This corresponds to a weak policy performance under the DSA framework.

BACKGROUND AND RECENT DEVELOPMENTS

1. Chad's external public debt burden increased considerably from 2008, mainly on account of external commercial borrowings related to oil. At March 2018, outstanding public and publicly guaranteed (PPG) external debt stood at nearly US$2.86 billion (26 percent of GDP) compared to US$1.6 billion (18 percent of GDP) at end-2008 (Text Table 1). Commercial borrowings (oil sale advances) from Glencore in 2013 to cover revenue shortfalls and in 2014 to purchase a share in the Doba oil Consortium were the main contributors. Falling oil prices over 2014-16 also contributed to the rising debt service burden by reducing revenues available to repay oil sales advances.

Text Table 1. Chad: External Debt Stock 2014-End-March 2018 /1

	2014	2015	2016	2017	2018 end-March
Total (Millions of US$)	3.8	2.7	2.6	2.8	2.9
(Billions of CFA francs)	2010	1617	1622	1572	1540
(Percent of GDP)	29	25	27	27	26
Billions of CFA francs					
Multilateral	735	375	390	385	405
IMF	11	38	77	96	122
World Bank/IDA	397	113	110	101	99
African Development Fund/Bank	181	69	56	56	55
Others	146	155	147	133	129
Bilateral	334	366	370	419	396
Paris Club official debt	11	2	...	25	24
Non-Paris Club official debt	323	364	370	394	372
of which: China, People's Republic	129	144	156	132	132
Libya	140	158	164	150	127
India	21	27	30	27	27
Commercial 2/	941	875	862	768	740
Share of Total (percent)					
Multilateral	37	23	24	24	26
Bilateral	17	23	23	27	26
Commercial 2/	47	54	53	49	48

Sources: Chadian authorities, selected creditors, and World Bank and IMF staff estimates.

[1]Includes only debt denominated in foreign currency.

[2]Glencore loan accounts for about 98 percent of commercial debt stock in 2017.

2. Debt relief, following the achievement of HIPC completion point, along with some debt reprofiling helped to ease the rising debt burden. Chad benefited from US$756 million in debt relief after achieving the HIPC completion point in April 2015. This amount includes MDRI relief from the International Development Association and the African Development Bank, and forgiveness from the Paris Club, while non-Paris Club members agreed to reschedule their remaining amounts on IDA comparable terms. In late 2015, the authorities also signed a rescheduling agreement with Glencore to consolidate the oil sale advances and extend their maturities. However, while the rescheduling agreement provided some flow relief, it proved to be insufficient, and led to an

increase in the present value of the debt. In February 2018, the authorities reached an agreement in principle with Glencore for a deeper restructuring which helped to reestablish debt sustainability. In April 2017, the authorities rescheduled arrears (accrued in 2016) and upcoming maturities with China.

3. The composition of external public debt has changed significantly over the past decade. The share of external debt from multilaterals has fallen sharply from about 86.5 percent in 2008 to 24.1 percent in 2017, while the share of commercial debt, which was virtually non-existent in 2008, has risen to almost 50 percent, mostly to Glencore. Bilateral debt doubled over the decade but, as a share of total debt, it is still significantly less than commercial debt (Text Table 1).

4. External payment arrears accumulated in 2016 and in 2017. As a result of liquidity challenges in 2016 and the first half of 2017, the government accrued external arrears vis-à-vis a number of multilateral, bilateral, and one commercial creditor (a bank from Taiwan province of China). At end-2017 about US$102 million (1 percent of GDP) remained outstanding, mainly to bilateral creditors (Text Table 2). The authorities have since reduced this stock to US$55 million by paying the amount owed to the Islamic Development Bank and through a rescheduling agreement with Libya. The authorities are making an effort to address remaining arrears. In early 2018, temporary arrears accumulated, but a portion was repaid after relatively short delays.

Text Table 2. Chad: Estimated External Arrears
(Millions of US dollars)

	2017	2018 (end-April)
Multilateral	**6.7**	**0.2**
Islamic Development Bank	6.5	0.0
European Investment Bank	0.2	0.2
International Fund for Agriculture Development		
Bilateral	**92.0**	**51.7**
Libya	40.3	0.0
India	10.2	10.2
Congo[1]	37.9	37.9
Equatorial Guinea	3.5	3.5
Commercial	**3.2**	**3.2**
Mega International Commercial Bank[2]	3.2	3.2
Total	**101.9**	**55.1**

[1] In CFAF
[2] Commercial bank from Taiwan Province of China.

5. Domestic public debt, which includes external debt denominated in domestic currency, has increased significantly in recent years. There has been a greater reliance on non-central bank financing, in particular issuance of government securities. While debt to the regional central bank (BEAC) remains high, its share in total debt has declined. In addition, in September 2017, all debt to the BEAC (for an amount of CFAF 479 billion) was consolidated and rescheduled into long-term securities with grace period of 4 years and maturity of 14 years (Text Table 3). In April 2018, domestic public debt rose by about 1 percent of GDP (CFAF 55 billion) as the government

cleared the arrears of the cotton public enterprise to commercial banks. This will likely be more than fully offset by the ongoing efforts to reduce existing debt to banks.

Glencore Debt Restructuring

6. The authorities have signed the final debt restructuring agreement with Glencore in June 2018 which will enter into effect as soon as all remaining standard procedural requirements in the agreement are satisfied shortly. The restructuring reestablishes debt sustainability and alleviates budgetary pressures. The agreement includes a significantly longer maturity, lower interest rate and fees, and contingency mechanisms to adjust debt service depending on oil revenues – which help maintain debt sustainability under different price and production scenarios (see below for further discussion).

Text Table 3. Chad: Domestic Debt at Year-End, 2014-2017

	2014	2015	2016	2017p
Total (Billions of CFA francs)	**708.9**	**1185.0**	**1482.2**	**1445.6**
(Percent of GDP)	**10.3**	**18.3**	**24.8**	**25.2**
Share of Total (in percent)				
Central Bank financing	31.4	38.4	33.3	33.2
Statutory advances [1]	*26.5*	*23.6*	*18.9*	...
Exceptional advance [1]		*11.8*	*11.5*	...
Consolidated debt	*4.9*	*2.9*	*3.0*	...
Commercial banks' loans	19.4	7.3	3.3	3.6
2011 Bond[2]	7.6	2.3	0.0	0.0
2013 Bond[2]	10.2	4.6	3.7	1.2
Treasury Bonds[3]		11.8	21.2	21.8
BDEAC	1.7	1.7	3.2	3.4
Republic of Congo	4.9	3.0	2.4	2.4
Equatorial Guinea	2.1	1.3	1.0	1.0
Cameroon			2.0	2.1
Domestic arrears[4]	7.3	16.9	12.8	13.5
Others[5]	12.4	7.4	5.9	6.1
Memo items:				
Treasury Bills	*3.9*	*7.1*	*11.2*	*11.7*

Source: Chadian authorities

[1]Existing balances were converted into long-term securities with grace period of 4 years and maturity of 14 years.

[2]Issued through banks' syndication

[3]Auctionned in regional securities' market.

[4]Assumes repayment of 20 billion in recognized arrears since June 2017 bring the total for 2017 to 51 billion.

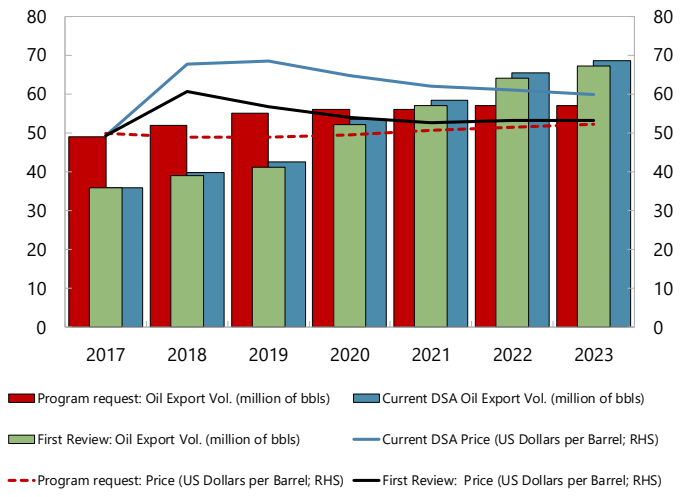
[5]Legal commitments, Standing payment orders, and accounting arrears.

UNDERLYING ASSUMPTIONS

7. The DSA's baseline scenario reflects policies and financing assumptions underlying the ECF arrangement and the Glencore debt restructuring. It is based on an upwardly revised WEO oil price projection (April 2018), a gradual recovery in oil production, conservative interest rate (LIBOR) projections, and policies to stabilize the fiscal position and support a sustainable recovery in non-oil activity. It also assumes clearance of external arrears in 2018.

- **Oil exports**. Chad's medium- and long-term macroeconomic outlook is characterized by a gradual increase in oil production. In 2017, oil production was significantly lower than projected at the time of the program request due to technical problems faced by the second largest oil producer in Chad. Production is expected to begin to recover in 2018 but delays in implementing new extraction technologies will keep production below the program request DSA projections up to 2020 (but slightly higher than at the first review), after which production is expected to increase gradually in line with the use of this technology and with the capacity of new fields projected to start production (Text Figure 1). The price of a barrel of Chadian oil has been revised upward in the medium-term relative to both the program request and first review DSAs projections reflecting a significant decline in the discount applied to Chad oil in the past year and higher WEO world oil price projections, leading also to slightly higher GDP growth.

Text Figure 1. Chad: Changes in Oil Projections



- **Fiscal policy.** It is assumed that the authorities remain committed to strengthen the fiscal position, including by maintaining the tight spending control, and improving non-oil revenue mobilization.

- **Arrears.** Efforts are underway to clear arrears to official bilateral creditors and the authorities are making good faith efforts to clear arrears to the bank from Taiwan province of China. The baseline scenario also includes a gradual reduction in the stock of verified domestic arrears and a start of modest payment of yet to be audited arrears.

- **Current account.** The current account deficit is now smaller in the medium-term than under the first review DSA mainly reflecting higher oil prices.

8. Risks to the outlook. The outlook is subject to a number of risks, largely downside in nature, including the potential for additional domestic debt and arrears not identified yet, a rise in non-concessional borrowing, and overruns in the wage bill. In addition, a further deterioration in the liquidity position of banks presents a risk given that it could undermine the rollover of domestic public debt. Developments in the international oil market continue to pose both upside and downside risks to the outlook, although the contingencies integrated into the Glencore debt restructuring agreement dampen the impact of fluctuations in oil prices on the fiscal position.

Text Table 4. Chad: DSA Assumptions

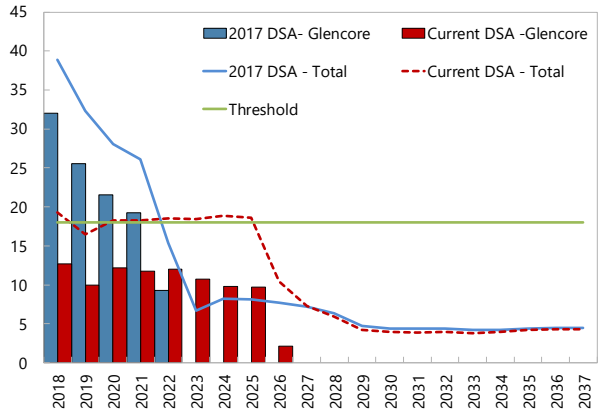
	2017	2018	2019	2020	2021	2022
			(Percentage Change)			
Real Growth						
Program request	0.6	2.4	3.1	3.9	3.6	3.7
First Review	-3.1	3.5	2.8	6.8	4.8	4.6
Current DSA	-3.1	3.5	3.6	6.9	4.8	4.7
GDP deflator in US dollar terms						
Program request	-4.1	1.5	2.3	3.2	2.7	2.3
First Review	1.0	12.4	3.6	4.1	3.7	3.6
Current DSA	1.0	8.7	2.1	4.1	3.4	3.6
			(Percent of GDP)			
Budget Balance						
Program request	3.3	2.7	2.1	2.3	2.1	2.4
First Review	0.6	2.1	0.9	1.7	1.5	2.3
Current DSA	1.5	2.8	1.8	2.9	2.6	3.3
Net FDI						
Program request	3.2	3.4	3.7	4.0	4.0	3.6
First Review	3.7	4.0	4.4	4.6	4.7	3.5
Current DSA	3.7	4.0	4.3	4.5	4.6	3.5
Non-Interest Current Account Balance						
Program request	-0.8	-1.9	-2.5	-2.7	-3.3	-2.7
First Review	-4.2	-3.6	-4.8	-4.2	-4.3	-4.0
Current DSA	-4.6	-0.3	-1.9	-2.2	-3.4	-3.8

EXTERNAL DSA

9. The Glencore debt restructuring has led to a significant improvement in the debt dynamics, particularly for the debt service-to-revenue ratio. The impact of the restructuring of the Glencore debt on debt dynamics was assessed in detail at the time of the first review DSA. The restructuring reduces significantly the debt service to revenue ratio which was mainly responsible for the debt difficulties Chad has faced recently. The ratio drops considerably over the next four years compared to the pre-Glencore restructuring (see Text Figure 2). The main changes relative to the

first review DSA is that the current DSA shows that (i) the debt service to revenue ratio is above the threshold in 2018 because during the restructuring negotiations the authorities delayed a cargo shipment due to Glencore (that is intended primarily to service the debt) in December 2017 to 2018; and (ii) after falling below threshold in 2019 to 16.5 percent (around half of the pre-restructuring ratio), it increases slightly over the medium term, marginally breaching the threshold before falling significantly because the adjustment mechanism in the new Glencore agreement requires the authorities to pay down debt faster when oil prices are higher than the contract baseline price. Under the new oil prices scenario, the cash sweep mechanism is triggered throughout the projection period and until 2026. This includes the payment of additional amortization and additional interest, and results in faster repayment of the debt which leads to a faster decline in the PV of debt (relative to the previous projection).

Text Figure 2. Chad: Debt Service-To-Revenue
(In percent)



10. All other external debt burden indicators are significantly below their respective thresholds under the baseline. The PV of public and publicly guaranteed external debt as a share of GDP declines gradually from 25.1 percent at end-2017 to under 10 percent by 2025 (Figure 1; Table 1). The PV of debt-to-revenue ratio is 8.7 percentage points below the 200 percent threshold mark in 2018 and declines over the medium-term. And the PV of debt to exports, and debt service to export measures remain comfortably below threshold over the span of the forecast horizon.

11. All the debt indicators breach the relevant thresholds in the presence of extreme shocks (Figure 1, Table 2). A shock to exports would push the PV of debt-to-exports and the debt service-to-exports ratios well above their thresholds. The shock that generates the largest impact for the PV of debt-to-GDP ratio, the PV of debt-to-revenue ratio, and debt service-to-revenue ratio is a combination shock where both growth and the primary balance fall below their historical average by half a standard deviation. All three debt burden indicators would rise sharply above their respective threshold in 2018 and remain elevated well into the medium-term. This highlights the sensitivity of the debt trajectory to the fiscal and growth assumptions and further confirms the need to adhere to the fiscal adjustment path under the IMF-supported program.

PUBLIC DSA

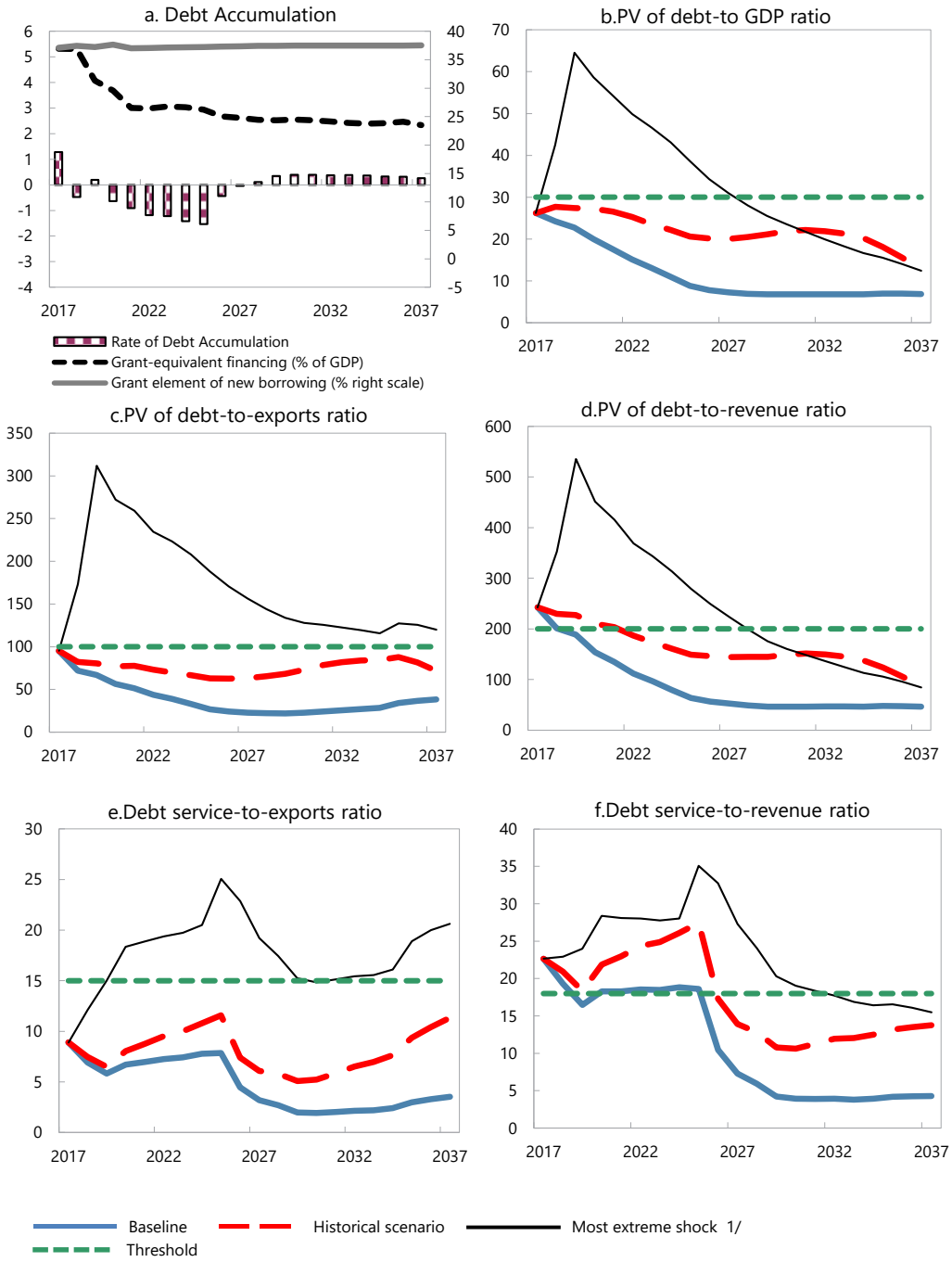
12. Analysis of total public debt suggests a heightened level of vulnerability (Figure 2, Table 3). The PV of total public debt, as a share of GDP, at end-2017 stood at 50.3 percent, which is about 12.1 percentage points above the benchmark level associated with heightened public debt vulnerabilities for weak policy performance. However, this indicator declines continuously over the

medium-term, falling below the threshold by 2020 and eventually stabilizing at about 17 percent into the long-term. The fixed primary balance scenario follows a similar trajectory but remains above the baseline, underscoring the need to maintain prudent fiscal policies (Figure 2).

CONCLUSION

13. Chad's external debt is assessed to be at high risk of distress and there are heightened public debt vulnerabilities. The rescheduling of the Glencore debt, along with the projected recovery in the oil sector and prudent fiscal policy, result in debt burden indicators declining significantly over the near and medium terms. The external debt trajectory remains sensitive to a number of shocks including on exports and to fiscal slippages. While progress has been made in clearing external arrears, temporary arrears continued to arise in the first half of 2018 suggesting that some difficulties in managing external debt remain. Chad's external debt is therefore assessed to be at high risk of debt distress. Additionally, total public debt vulnerabilities remain elevated, which reinforces the need to maintain prudent fiscal policy including on external and domestic borrowing. While progress has been made recently to reduce the stock of external and domestic arrears, much more attention is needed going forward to clear all the remaining arrears. Finally, effective inter-agency coordination to strengthen the capacity to record and monitor public debt is very important to better manage public debt. The authorities broadly agreed with staff's assessment of debt sustainability in Chad and have expressed their commitment to further reduce debt vulnerabilities going forward.

Figure 1. Chad: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a Combination shock

Figure 2. Chad: Indicators of Public Debt under Alternative Scenarios, 2017–37 1/

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Table 1. Chad: External Debt Sustainability Framework, Baseline Scenario, 2014–37 1/

(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**29.1**	**25.0**	**27.1**			**27.3**	**26.2**	**24.9**	**22.2**	**19.7**	**17.2**		**9.2**	**9.2**	
of which: public and publicly guaranteed (PPG)	29.1	25.0	27.1			27.3	26.2	24.9	22.2	19.7	17.2		9.2	9.2	
Change in external debt	8.0	-4.1	2.2			0.2	-1.2	-1.3	-2.7	-2.5	-2.5		-0.5	-0.1	
Identified net debt-creating flows	2.2	16.7	8.9			2.9	-1.1	-0.4	-1.7	-1.0	-0.2		-0.2	3.5	
Non-interest current account deficit	**8.2**	**12.5**	**7.2**	**5.3**	**6.3**	**4.6**	**3.1**	**4.6**	**3.6**	**4.0**	**3.5**		**1.9**	**4.6**	2.5
Deficit in balance of goods and services	12.5	16.3	15.0			13.9	10.6	10.5	8.8	8.8	8.0		4.7	8.7	
Exports	31.4	26.5	24.4			27.5	33.7	34.1	35.3	34.2	34.5		31.7	17.9	
Imports	43.9	42.9	39.4			41.4	44.3	44.5	44.1	43.0	42.5		36.4	26.6	
Net current transfers (negative = inflow)	-7.9	-7.1	-7.7	-6.1	1.5	-9.6	-9.5	-8.1	-7.6	-6.6	-6.4		-5.1	-4.1	-4.8
of which: official	-4.4	-3.0	-2.9			-3.6	-4.3	-3.0	-2.6	-1.9	-1.9		-1.5	-1.6	
Other current account flows (negative = net inflow)	3.6	3.2	-0.2			0.2	1.9	2.3	2.3	1.8	2.0		2.3	0.0	
Net FDI (negative = inflow)	**-5.2**	**-5.1**	**-2.4**	**-4.3**	**1.9**	**-3.7**	**-4.3**	**-5.0**	**-4.5**	**-4.6**	**-3.5**		**-2.0**	**-1.0**	-1.7
Endogenous debt dynamics 2/	**-0.8**	**9.3**	**4.2**			**2.0**	**0.2**	**0.0**	**-0.7**	**-0.3**	**-0.3**		**-0.2**	**-0.1**	
Contribution from nominal interest rate	0.7	1.2	2.1			1.1	1.0	0.9	0.8	0.7	0.6		0.1	0.1	
Contribution from real GDP growth	-1.4	-0.7	1.7			0.9	-0.9	-0.9	-1.5	-1.0	-0.8		-0.4	-0.2	
Contribution from price and exchange rate changes	-0.2	8.8	0.4			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**5.7**	**-20.8**	**-6.7**			**-2.7**	**-0.1**	**-0.8**	**-1.0**	**-1.5**	**-2.2**		**-0.3**	**-3.7**	
of which: exceptional financing	-0.1	-0.8	-1.1			-1.0	-1.8	-1.1	-0.7	-0.3	-0.3		-0.2	-0.1	
PV of external debt 4/	...	...	26.9			26.2	24.3	22.8	20.0	17.6	15.1		7.3	6.9	
In percent of exports	...	...	110.4			95.3	72.0	66.8	56.6	51.3	43.8		22.9	38.3	
PV of PPG external debt	**...**	**...**	**26.9**			**26.2**	**24.3**	**22.8**	**20.0**	**17.6**	**15.1**		**7.3**	**6.9**	
In percent of exports	**...**	**...**	**110.4**			**95.3**	**72.0**	**66.8**	**56.6**	**51.3**	**43.8**		**22.9**	**38.3**	
In percent of government revenues	**...**	**...**	**279.5**			**242.6**	**201.3**	**188.9**	**153.9**	**134.6**	**112.0**		**52.4**	**46.5**	
Debt service-to-exports ratio (in percent)	**15.6**	**9.5**	**16.7**			**8.9**	**6.9**	**5.8**	**6.7**	**7.0**	**7.3**		**3.2**	**3.5**	
PPG debt service-to-exports ratio (in percent)	**15.6**	**9.5**	**16.7**			**8.9**	**6.9**	**5.8**	**6.7**	**7.0**	**7.3**		**3.2**	**3.5**	
PPG debt service-to-revenue ratio (in percent)	**29.9**	**24.0**	**42.2**			**22.6**	**19.3**	**16.5**	**18.3**	**18.3**	**18.5**		**7.3**	**4.3**	
Total gross financing need (Billions of U.S. dollars)	1.1	1.1	0.9			0.3	0.1	0.2	0.2	0.2	0.4		0.2	1.5	
Non-interest current account deficit that stabilizes debt ratio	0.2	16.6	5.0			4.4	4.2	5.9	6.3	6.5	6.0		2.5	4.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.9	1.8	-6.4	4.1	5.3	-3.1	3.5	3.6	6.9	4.8	4.6	3.4	3.8	2.8	2.9
GDP deflator in US dollar terms (change in percent)	0.8	-23.2	-1.5	-0.1	12.5	1.0	8.7	2.1	4.1	3.4	3.6	3.8	2.9	2.9	2.8
Effective interest rate (percent) 5/	3.6	3.3	7.6	2.7	2.1	4.1	4.3	3.5	3.7	3.3	3.2	3.7	1.3	1.3	1.5
Growth of exports of G&S (US dollar terms, in percent)	1.3	-33.9	-15.3	-1.7	19.6	10.4	37.8	7.0	15.4	5.0	9.5	14.2	5.3	-0.7	1.4
Growth of imports of G&S (US dollar terms, in percent)	9.8	-23.6	-15.3	1.9	14.1	2.9	20.3	6.3	10.2	5.7	7.1	8.8	4.2	1.4	2.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	37.1	37.4	37.3	37.6	37.0	37.1	37.3	37.4	37.6	37.4
Government revenues (excluding grants, in percent of GDP)	16.4	10.5	9.6			10.8	12.0	12.0	13.0	13.0	13.5		13.9	14.7	14.3
Aid flows (in Billions of US dollars) 7/	0.4	0.4	0.3			0.6	0.6	0.5	0.5	0.4	0.5		0.6	0.9	
of which: Grants	0.3	0.4	0.3			0.4	0.5	0.4	0.4	0.4	0.4		0.5	0.7	
of which: Concessional loans	0.1	0.1	0.0			0.2	0.1	0.1	0.1	0.1	0.1		0.1	0.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.3	5.3	4.1	3.7	3.0	3.0		2.6	2.3	2.6
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			72.2	77.0	79.8	82.6	82.9	83.3		81.1	80.8	80.9
Memorandum items:															
Nominal GDP (Billions of US dollars)	14.0	10.9	10.1			9.9	11.1	11.8	13.1	14.2	15.4		21.4	36.1	
Nominal dollar GDP growth	7.8	-21.8	-7.8			-2.2	12.6	5.8	11.3	8.4	8.4	7.4	6.8	5.8	5.9
PV of PPG external debt (in Billions of US dollars)			2.6			2.7	2.7	2.7	2.6	2.5	2.3		1.6	2.5	
(PVt-PVt-1)/GDPt-1 (in percent)						**1.3**	-0.5	0.2	-0.6	-0.9	-1.2	-0.3	0.0	0.3	-0.1
Gross workers' remittances (Billions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	26.9			26.2	24.3	22.8	20.0	17.6	15.1		7.3	6.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	110.4			95.3	72.0	66.8	56.6	51.3	43.8		22.9	38.3	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	16.7			8.9	6.9	5.8	6.7	7.0	7.3		3.2	3.5	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief), changes in gross foreign assets, and valuation adjustments. Projections also include contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37

(Percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	25	23	22	19	17	14	**7**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	25	28	29	30	29	28	**24**	15
A2. New public sector loans on less favorable terms in 2017-2037 2/	25	23	23	20	18	16	**9**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	25	24	24	21	18	16	**8**	8
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	25	32	44	39	36	33	**21**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	25	28	32	28	24	21	**11**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	25	27	29	26	23	21	**12**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	25	41	62	56	52	48	**30**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	25	32	30	26	23	20	**10**	9
PV of debt-to-exports ratio								
Baseline	91	68	64	54	48	41	**23**	38
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	91	83	85	84	86	81	**75**	84
A2. New public sector loans on less favorable terms in 2017-2037 2/	91	69	66	56	52	45	**30**	60
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	91	68	64	54	49	41	**23**	38
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	91	167	304	265	252	229	**156**	120
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	91	68	64	54	49	41	**23**	38
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	91	80	86	74	68	60	**37**	42
B5. Combination of B1-B4 using one-half standard deviation shocks	91	159	261	228	218	198	**137**	98
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	91	68	64	54	49	41	**23**	38
PV of debt-to-revenue ratio								
Baseline	232	191	180	146	127	105	**52**	46
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	232	232	241	229	225	207	**172**	102
A2. New public sector loans on less favorable terms in 2017-2037 2/	232	193	187	153	136	115	**68**	73
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	232	199	199	161	140	116	**58**	51
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	232	266	361	303	277	245	**150**	61
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	232	236	264	213	186	154	**76**	68
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	232	224	243	200	179	154	**86**	51
B5. Combination of B1-B4 using one-half standard deviation shocks	232	339	517	435	400	355	**219**	84
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	232	262	250	202	176	146	**72**	64

Table 2. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37 (concluded)

(Percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
Debt service-to-exports ratio								
Baseline	9	7	6	7	7	7	**3**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	9	7	6	8	9	10	**6**	11
A2. New public sector loans on less favorable terms in 2017-2037 2/	9	7	6	7	7	7	**3**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	9	7	6	7	7	7	**3**	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	9	12	15	18	19	19	**19**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	9	7	6	7	7	7	**3**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	9	7	6	7	7	8	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	9	11	12	15	15	16	**17**	18
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	9	7	6	7	7	7	**3**	4
Debt service-to-revenue ratio								
Baseline	23	19	16	18	18	19	**7**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	23	21	18	22	23	24	**14**	14
A2. New public sector loans on less favorable terms in 2017-2037 2/	23	19	16	18	19	19	**7**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	23	20	18	20	20	20	**8**	5
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	23	19	18	21	21	21	**18**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	23	24	24	27	27	27	**11**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	23	19	17	19	19	19	**11**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	23	23	24	28	28	28	**27**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	23	27	23	25	25	26	**10**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	36	36	36	36	36	36	**36**	36

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that interest rate on new borrowing is by 2 percentage points higher than in the baseline. Grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Chad: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014-2037

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	39.3	43.3	51.7			52.5	49.2	45.4	40.8	36.9	33.1	43.0	21.4	19.9	19.8
of which: foreign-currency denominated	29.1	25.0	27.1			27.3	26.2	24.9	22.2	19.7	17.2		9.2	9.2	
Change in public sector debt	8.8	3.9	8.5			0.8	-3.3	-3.8	-4.6	-3.9	-3.8		-1.1	-0.2	
Identified debt-creating flows	0.1	5.4	3.9			-2.6	-3.9	-4.0	-6.1	-4.9	-5.5		-4.1	-0.8	
Primary deficit 6/	2.6	3.0	-0.4	0.9	3.9	-1.2	-2.8	-1.8	-2.9	-2.6	-3.4	-2.4	-2.8	0.0	-1.9
Revenue and grants	18.4	14.0	12.6			14.9	16.4	15.5	16.2	15.7	16.2		16.1	16.7	
of which: grants	2.0	3.4	2.9			4.1	4.4	3.5	3.2	2.6	2.6		2.3	2.0	
Primary (noninterest) expenditure	21.0	17.0	12.2			13.7	13.6	13.8	13.3	13.1	12.8		13.3	16.7	
Automatic debt dynamics	1.9	3.0	4.8			0.4	-0.5	-1.8	-2.9	-2.0	-1.8		-1.1	-0.7	
Contribution from interest rate/growth differential	-1.0	-2.6	5.0			4.2	0.9	-1.7	-2.6	-1.8	-1.6		-1.1	-0.7	
of which: contribution from average real interest rate	1.0	-1.9	2.0			2.5	2.7	0.0	0.3	0.1	0.0		-0.2	-0.2	
of which: contribution from real GDP growth	-2.0	-0.7	3.0			1.7	-1.8	-1.7	-2.9	-1.9	-1.6		-0.8	-0.5	
Contribution from real exchange rate depreciation	2.9	5.6	-0.3			-3.8	-1.3	-0.1	-0.3	-0.2	-0.2		...	...	
Other identified debt-creating flows	-4.4	-0.7	-0.5			-1.8	-0.6	-0.4	-0.3	-0.3	-0.3		-0.2	-0.1	
Privatization receipts (negative)	-4.0	-0.4	0.0			-1.3	-0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.4	-0.3	-0.5			-0.5	-0.4	-0.4	-0.3	-0.3	-0.3		-0.2	-0.1	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	8.7	-1.4	4.6			3.4	0.6	0.2	1.5	1.0	1.7		3.0	0.7	
Other Sustainability Indicators															
PV of public sector debt	...	...	51.5			51.4	47.3	43.3	38.6	34.8	31.0		19.4	17.6	
of which: foreign-currency denominated	...	...	26.9			26.2	24.3	22.8	20.0	17.6	15.1		7.3	6.9	
of which: external	...	...	26.9			26.2	24.3	22.8	20.0	17.6	15.1		7.3	6.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.4	8.2	6.2			5.8	5.9	5.2	5.3	4.0	3.1		2.0	4.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	410.2			344.4	288.3	279.0	238.3	221.8	192.0		120.4	105.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	535.1			475.3	392.6	359.1	297.4	266.7	229.5		140.0	119.2	
of which: external 3/	...	...	279.5			242.6	201.3	188.9	153.9	134.6	112.0		52.4	46.5	
Debt service-to-revenue and grants ratio (in percent) 4/	31.6	33.6	41.7			27.3	36.2	23.5	33.5	25.3	24.2		14.7	12.5	
Debt service-to-revenue ratio (in percent) 4/	35.3	44.6	54.4			37.6	49.3	30.2	41.8	30.4	29.0		17.1	14.2	
Primary deficit that stabilizes the debt-to-GDP ratio	-6.2	-0.9	-8.9			-2.0	0.4	2.1	1.7	1.3	0.4		-1.7	0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.9	1.8	-6.4	4.1	5.3	-3.1	3.5	3.6	6.9	4.8	4.6	3.4	3.8	2.8	2.9
Average nominal interest rate on forex debt (in percent)	3.6	3.3	7.6	2.7	2.1	4.1	4.3	3.5	3.7	3.3	3.2	3.7	1.3	1.3	1.5
Average real interest rate on domestic debt (in percent)	5.1	-1.5	-1.3	1.7	3.4	2.2	1.6	-0.6	-0.5	-0.4	-0.9	0.2	-0.7	-0.4	-0.5
Real exchange rate depreciation (in percent, + indicates depreciation)	14.3	20.8	-0.9	3.7	13.9	-12.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-2.0	3.3	3.1	1.2	3.6	-0.7	0.3	2.2	2.6	2.7	2.9	1.7	2.8	3.0	2.9
Growth of real primary spending (deflated by GDP deflator, in percent)	4.8	-26.6	-35.9	-5.7	13.7	8.9	4.5	5.2	4.1	3.2	2.0	4.7	5.4	3.5	4.8
Grant element of new external borrowing (in percent)	...	...	...	...	...	37.1	37.4	37.3	37.6	37.0	37.1	37.3	37.4	37.6	...

Sources: Country authorities; and staff estimates and projections.

1/ The coverage of public sector debt comprises the obligations of the central government, including commercial debt. The definition of debt corresponds to gross debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

6/ The primary deficit grosses up oil revenue and debt service on the oil sales advances

Table 4. Chad: Current Policies: Sensitivity Analysis for Key Indicators of Public Debt, 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	51	47	43	39	35	31	19	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	51	49	47	45	44	42	41	40
A2. Primary balance is unchanged from 2017	51	48	45	41	38	35	29	26
A3. Permanently lower GDP growth 1/	51	48	45	41	38	35	29	48
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	51	50	49	45	41	38	30	37
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	51	52	52	47	43	39	27	25
B3. Combination of B1-B2 using one half standard deviation shocks	51	52	51	47	43	39	29	30
B4. One-time 30 percent real depreciation in 2018	51	58	54	48	44	39	26	24
B5. 10 percent of GDP increase in other debt-creating flows in 2018	51	54	49	45	41	37	25	24
PV of Debt-to-Revenue Ratio 2/								
Baseline	344	288	279	238	222	192	120	105
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	344	299	300	277	276	260	253	245
A2. Primary balance is unchanged from 2017	344	293	286	251	240	217	179	152
A3. Permanently lower GDP growth 1/	344	290	285	249	238	213	173	278
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	344	299	307	269	259	232	182	216
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	344	315	333	288	272	241	167	148
B3. Combination of B1-B2 using one half standard deviation shocks	344	312	327	285	271	242	178	180
B4. One-time 30 percent real depreciation in 2018	344	353	344	295	276	242	159	141
B5. 10 percent of GDP increase in other debt-creating flows in 2018	344	325	318	274	259	228	156	141
Debt Service-to-Revenue Ratio 2/								
Baseline	27	36	23	33	25	24	15	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	27	37	24	35	27	26	19	23
A2. Primary balance is unchanged from 2017	27	37	24	34	26	25	17	18
A3. Permanently lower GDP growth 1/	27	37	24	35	27	26	17	22
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	27	38	26	37	28	27	18	20
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	27	37	24	36	27	26	19	17
B3. Combination of B1-B2 using one half standard deviation shocks	27	37	25	36	28	26	19	18
B4. One-time 30 percent real depreciation in 2018	27	40	29	41	33	32	19	18
B5. 10 percent of GDP increase in other debt-creating flows in 2018	27	37	25	35	27	25	18	16

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.